

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

May 2012

CL203 Fluid Mechanics

Date: 03.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define Density, Viscosity, Pressure and Vacuum pressure [04]
(b) One liter of crude oil weighs 9.6 N. Calculate its specific weight, density and specific gravity. [03]
- Q - 2 (a) Define Buoyancy, Center of buoyancy, Uniform flows and Laminar flows [04]
(b) What are the limitations of piezometers? [02]
(c) A vertical isosceles (two sides are same) triangle gate with its vertex up has a base width 1 m and height 1.5 m. If the vertex of gate is 1 m below the free water surface, find (i) total pressure force (ii) center of pressure [03]
(c) For a two dimensional potential flow, the velocity potential is given by $\phi = x(2y - 1)$ [05]
determine the velocity at the point P(4,5). Also determine the value of stream function Ψ at the point P.

OR

- Q - 2 (a) Explain the following terms: [03]
Stream function, Flow nets and Local acceleration
(b) State and prove hydrostatic law. [04]
(c) A micro manometer having a ratio of reservoir to limbs areas as 100 was used to determine the pressure in pipe containing liquid of specific gravity 0.8. Determine the pressure in the pipe for manometer reading as shown in Figure 1. [03]
(d) An open circular cylinder of 15 cm diameter and 100 cm long contains water up to a height of 70 cm. Find the speed at which the cylinder is to be rotated about its vertical axis so that the axial depth becomes zero. [04]
- Q - 3 (a) The velocity components in a two dimensional flow are $u = y^3/3 + 2x - x^2y$ and $v = xy^2 - 2y - x^3/3$. Show that these components represent a possible case of irrotational flow. [05]

- (b) Give the definition and detailed classification of notches. [05]
- (c) A horizontal venturimeter with inlet diameter 150 mm and throat diameter 75 mm is used to measure the discharge of water. The differential manometer connected to the inlet gives reading of 150 mm of mercury. Determine the rate of flow if the coefficient of discharge is 0.98. [04]

OR

- Q - 3 (a) Derive an expression for continuity equation for three dimensional flow. [05]
- (b) A vertical sharp – edge orifice of 140 mm diameter is discharging water at the rate of 100 liters/s under a constant head of 100 m. A point on the jet measured from the vena contracta of the jet has coordinate 4.3 m horizontal and 0.6 m vertical. Determine the coefficients C_v , C_c and C_d of the orifice. [05]
- (c) An internal mouthpiece of 60 mm diameter is fitted to the side of a tank. It is discharging water under a constant head of 5 meters. Find the discharge through mouthpiece when (i) the mouthpiece running free and (ii) the mouthpiece running full. [04]

SECTION – II

- Q - 4 (a) Give the application of momentum equation. [02]
- (b) Water is flowing through a pipe having diameter 300 mm and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm² and pressure at the upper end is 9.81 N/cm². Determine the difference in the datum head if the rate of flow through pipe is 40 liters/s. [05]
- Q - 5 (a) Determine the head loss due to friction in a pipe of diameter 250 mm and length 60 m through which water is flowing at a velocity of 2.5 m/s using (i) Darcy's formula (ii) Chezy's formula for which $C = 60$. Assume kinematic viscosity for water as 0.01 stokes. [05]
- (b) Describe procedure to conduct the dimensional analysis using Rayleigh's method. [05]
- (c) A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.90 and viscosity 3×10^{-2} poise at the rate of 3000 liters/s. Tests were conducted on a 15 cm diameter pipe using water at 20 °C. Find the velocity and rate of flow in the model. [04]
- Viscosity of water at 20 °C = 0.01 poise.

OR

- Q - 5 (a) Briefly describe geometric similarity. [02]
- (b) Explain hydraulic gradient line and total energy line with neat sketch. [05]
- (c) The efficiency η of a fan depends on the density ρ , the dynamic viscosity μ of the [07]

fluid, the angular velocity ω , diameter D of the rotor and the discharge Q . Express η in terms of dimensionless parameter.

- Q - 6 (a) Describe in detail the energy thickness (δ^{**}). [03]
- (b) A fluid of viscosity 0.7 Ns/m^2 and specific gravity 1.3 is flowing through a circular pipe of diameter 100 mm. The maximum shear stress at the pipe wall is given as 196.2 N/m^2 . Find (i) the pressure gradient, (ii) the average velocity and (iii) Reynolds number of the flow. [05]
- (c) Enlist basic thermodynamic process. Derive Bernoulli's energy equation for one-dimensional compressible isothermal process. [06]

OR

- Q - 6 (a) Give the equation for stagnation density, stagnation temperature and Hagen-Poiseuille's formula. [03]
- (b) Calculate the velocity and Mach number of a supersonic aircraft flying at an altitude of 1200 m where the temperature is 0°C . The sound of the aircraft is heard 3 seconds after passage of aircraft over the head of an observer. Assume $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$ [04]
- (c) An oil of viscosity 0.1 Ns/m^2 and relative density 0.9 is passing through a circular pipe of diameter 50 mm and of length 300 m. The rate of flow of fluid through the pipe is 3.5 liters/s. Find the pressure drop in a length of 300 m and also the shear stress at the pipe wall. [07]

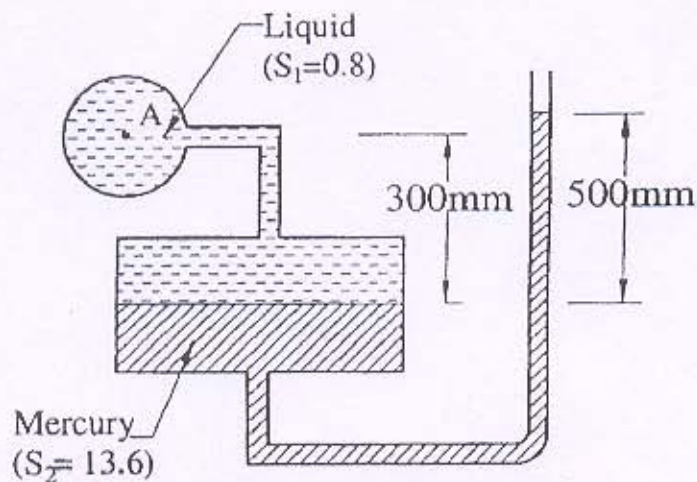


Figure: 1
